

Brief solutions/Answers to problems of Problem Set 8

1/ $P(\text{actually a spam}) = P(A_S) = 0.5$ (given prior prob.)

$P(\text{detected/learned as spam by the software}) = P(D_S) = ?$

Also given are the conditional probs.

$P(D_S|A_S) = 0.99$, $P(D_S|A_S^c) = 0.05$, to find $P(A_S^c|D_S)$
(note the corrected language of the problem)

$$P(A_S^c|D_S) = \frac{5}{104}.$$

2/ $X = \# \text{ successes} \sim \text{Bin}(7, \theta)$

$\theta \sim \text{Beta}(2, 2)$

$\therefore$ posterior distribution $\pi(\theta|x) := \text{Beta}(2+x, 7-x+2)$.

$\therefore \hat{\theta}_{\text{prior}} = 0.5$, $\hat{\theta}_{\text{posterior}} (= \hat{\theta}_{\text{Bayes}}) = \frac{9}{14} \approx 0.643$.

3/ Given, $x|\beta \sim \text{Gamma}(\alpha, \frac{1}{\beta})$, $\beta \sim \text{Gamma}(\lambda, \frac{1}{\gamma})$ (see corrected version).

Then conditional $f(x|\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} e^{-\beta x} x^{\alpha-1}$, prior $\pi(\beta) = \frac{\gamma^\lambda}{\Gamma(\lambda)} e^{-\gamma\beta} \beta^{\lambda-1}$

$\Rightarrow \pi(\beta|x) := \text{Gamma}(\alpha+\lambda, \frac{1}{x+\gamma})$, hence conjugate prior.

4. $x_1, \dots, x_n$ iid from Neg. Bin. (r, θ) , $x = \{0, 1, \dots\}$.

the conditional, $f(x|\theta) = \prod_{i=1}^n \binom{r+x_i-1}{x_i} \theta^r (1-\theta)^{x_i}$

the prior, $\pi(\theta) = \frac{1}{\text{Beta}(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

the posterior $\pi(\theta|x) = \frac{1}{B(r+\alpha, \sum x_i + \beta)} \theta^{r+\alpha-1} (1-\theta)^{\sum x_i + \beta - 1}$

$\therefore$ conjugate prior.

5. $f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$, $x=0,1,\dots,n$

2/15/8

$$\pi(\theta) = \frac{1}{\theta(1-\theta)}$$

$$\pi(\theta|x) = \frac{1}{\text{Beta}(x, n-x)} \theta^{x-1} (1-\theta)^{n-x-1} := \text{Beta}^-(x, n-x).$$

if $x > 0$ and $x < n$

$$\left. \begin{array}{l} \text{If } x=0, \pi(\theta|x) = \frac{(1-\theta)^{n-1}}{\theta} \\ x=n, \pi(\theta|x) = \frac{\theta^{n-1}}{1-\theta} \end{array} \right\} \text{ are improper.}$$

6. $f(x|\theta) = \frac{\theta^\alpha}{x!} e^{-n\theta} \frac{\theta^{\sum x_i}}{\prod_{i=1}^n x_i!}$, $\pi(\theta) = \frac{1}{\theta}$,

$$\pi(\theta|x) = \frac{1}{\prod x_i \left(\frac{1}{n}\right)^{\sum x_i}} e^{-n\theta} \theta^{\sum x_i - 1} := \text{Gamma}(\sum x_i, \frac{1}{n}).$$

$\therefore$ proper if $\sum x_i > 0$
improper $\sum x_i = 0$.

7. $f(x|\theta) = \frac{1}{\theta}$, $0 \leq x \leq \theta$

$$\pi(\theta) = \beta^2 \theta e^{-\beta \theta}$$

$$\pi(\theta|x) = \beta e^{-\beta(\theta-x)}, \quad 0 \leq x \leq \theta.$$

$$\therefore E(\theta|x=x) = \frac{1}{\beta} + x, \quad \therefore \delta_{\text{Bayes}} (= \hat{\theta}_{\text{Bayes}}) = x + \frac{1}{\beta}.$$

8.

$$f(x|\theta) = \theta^n e^{-\theta \sum x_i}, \quad x_i > 0; \theta > 0$$

$$\pi(\theta) = \frac{1}{\Gamma(\alpha) \beta^\alpha} e^{-\theta/\beta} \theta^{\alpha-1}$$

$$\text{Then } \pi(\theta|x) := \text{Gamma}(n+\alpha, n\bar{x} + \frac{1}{\beta}).$$

$$\text{Then (a), } \delta_{\text{Bayes}} (= \hat{\theta}_{\text{Bayes}}) = \frac{n+\alpha-1}{n\bar{x} + 1/\beta}$$

$$(b) \delta_{\text{Bayes}} (= \hat{\theta}_{\text{Bayes}}) = \left(\frac{n\bar{x} + 1/\beta}{n\bar{x} + \frac{1}{\beta} + 1} \right)^{n+\alpha-1}$$

9.

$$X = \# \text{ defects, } X|\theta \sim P(\theta).$$

$$\theta \sim \text{Gamma}(\alpha=3, \beta=1) \quad (\text{shape \& rate}).$$

$$\theta|x \sim \text{Gamma}(n\bar{x} + \alpha, n + \beta).$$

$$\therefore \delta_{\text{Bayes}} (\text{estimate}) = \frac{16}{6}.$$

10. $X|\theta \sim N(\theta, 1)$, $\pi(\theta) := N(68, 1)$

3/RS8

$\theta|x \sim N(E(\theta|x), \text{Var}(\theta|x))$

$\delta_{\text{Bayes}} = E(\theta|x) = \frac{x n \bar{x} / \sigma^2 + \mu_0 / \sigma_0^2}{n / \sigma^2 + 1 / \sigma_0^2} = 69.07$

with, $n=5$, $\bar{x}=69.5$, $\sigma^2=2^2$, $\mu_0=68$, $\sigma_0^2=1^2$

11. $X \sim N(\theta, 1)$, $\pi(\theta) := N(0, \sigma^2)$

$\therefore \pi(\theta|x) := N\left(\frac{x\sigma^2}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2}\right)$

Bayes risk = $E_X E_{\theta|x} \left[\left[\theta - \frac{x\sigma^2}{1+\sigma^2} \right]^2 | X=x \right] = E_X \left(\frac{\sigma^2}{1+\sigma^2} \right) = \frac{\sigma^2}{1+\sigma^2}$

12. $f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$

$\Gamma(\theta) = \frac{n}{\theta(1-\theta)}$, $\therefore \pi(\theta) \propto \{\theta(1-\theta)\}^{1/2}$

$\pi(\theta|x) := \text{Beta}(x+1/2, n-x+1/2)$

$\delta_{\text{Bayes estimate}} = E(\theta|x) - E(\theta^2|x)$

(replace x by X in estimator)
 $\delta_{\text{Bayes}} = \frac{x+1/2}{n+1} - \left[\frac{(x+1/2)^2}{(n+1)^2} - \frac{(x+1/2)(n-x+1/2)}{(n+1)^2(n+2)} \right]$

13. $f(x|\theta) = \frac{1}{\sqrt{2\pi}} \exp -\frac{1}{2}(x-\theta)^2$, $\pi(\theta) = e^\theta$

$\pi(\theta|x) = N(x+1, 1)$

$\therefore \delta_{\text{Bayes}} = E(\theta|x) = x+1$

14. X_i a: ith response, $X_i \sim \text{Ber}(p)$, $p = \text{prop. of 'yes'}$
 $\sum X_i \sim \text{Bin}(n, p)$, $\pi(p) := \text{Beta}(\alpha, \beta)$, $\therefore \pi(p|x) := \text{Beta}(\alpha + \sum X_i, \beta + n - \sum X_i)$
 and $E(p|x) = \frac{\sum X_i + \alpha}{n + \alpha + \beta}$, $\text{Var}(p|x) = \frac{(\sum X_i + \alpha)(\beta + n - \sum X_i)}{(n + \alpha + \beta)^2} = \sigma^2$, say.

Using Chebyshev's ineq. Prob. $[|p - \mu| \leq \sigma t] \geq 1 - \frac{1}{t^2}$, putting μ, σ and $1 - \frac{1}{t^2} = 0.96$,

the reqd: 96% credible interval for p is $[0.40, 0.50]$

15. $\pi(\theta|x) := \text{Gamma}(n+1, \frac{1}{\sum x_i})$, $E(\theta|x) = \mu$, $\text{Var}(\theta|x) = \sigma^2$, say.

Then $\frac{\theta - \mu}{\sigma} \stackrel{\text{asy}}{\sim} N(0, 1)$, and 100(1- α)% credible interval (approximate)

is $E(\theta|x) \pm z_{\alpha/2} \sqrt{\text{Var}(\theta|x)}$